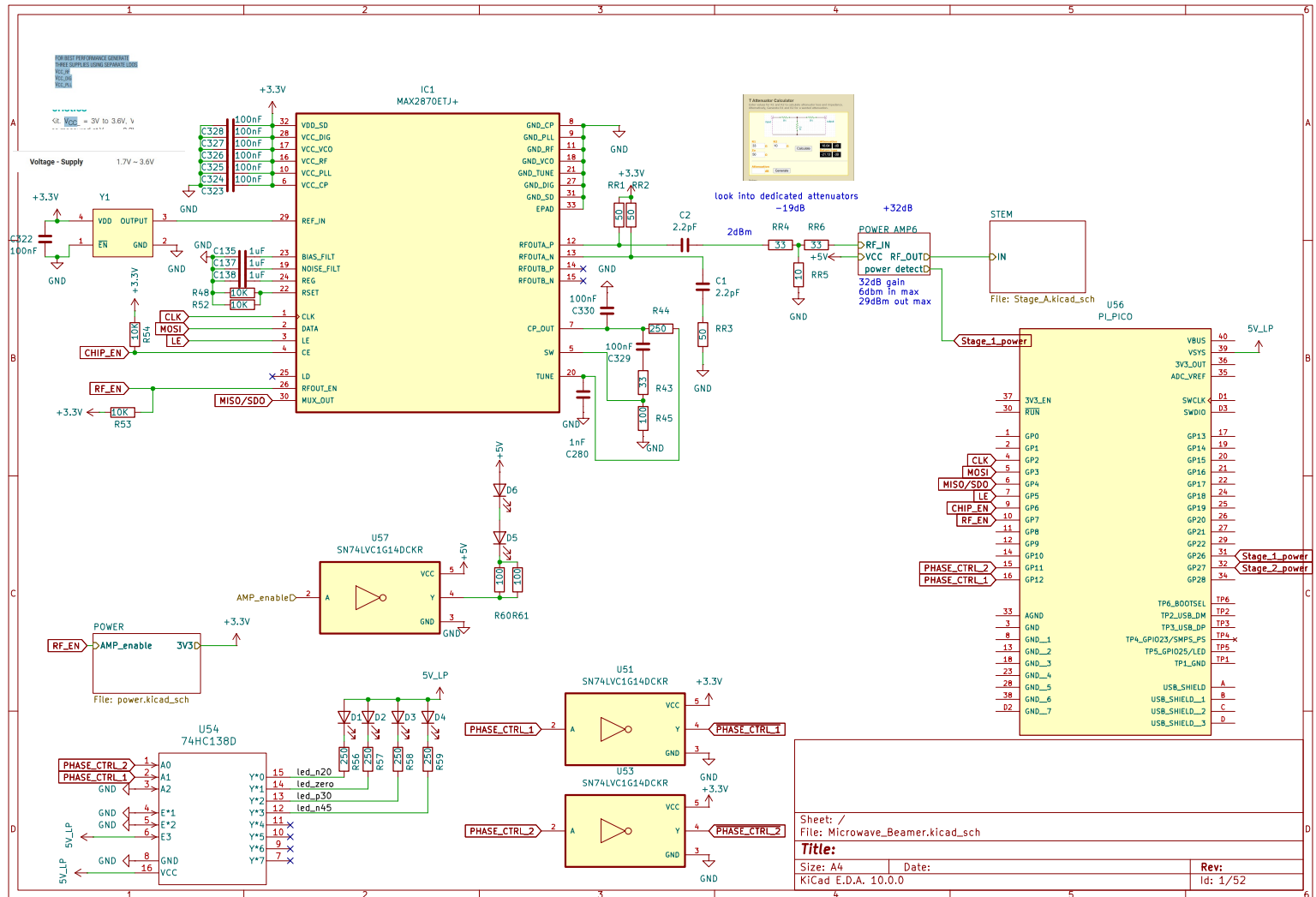
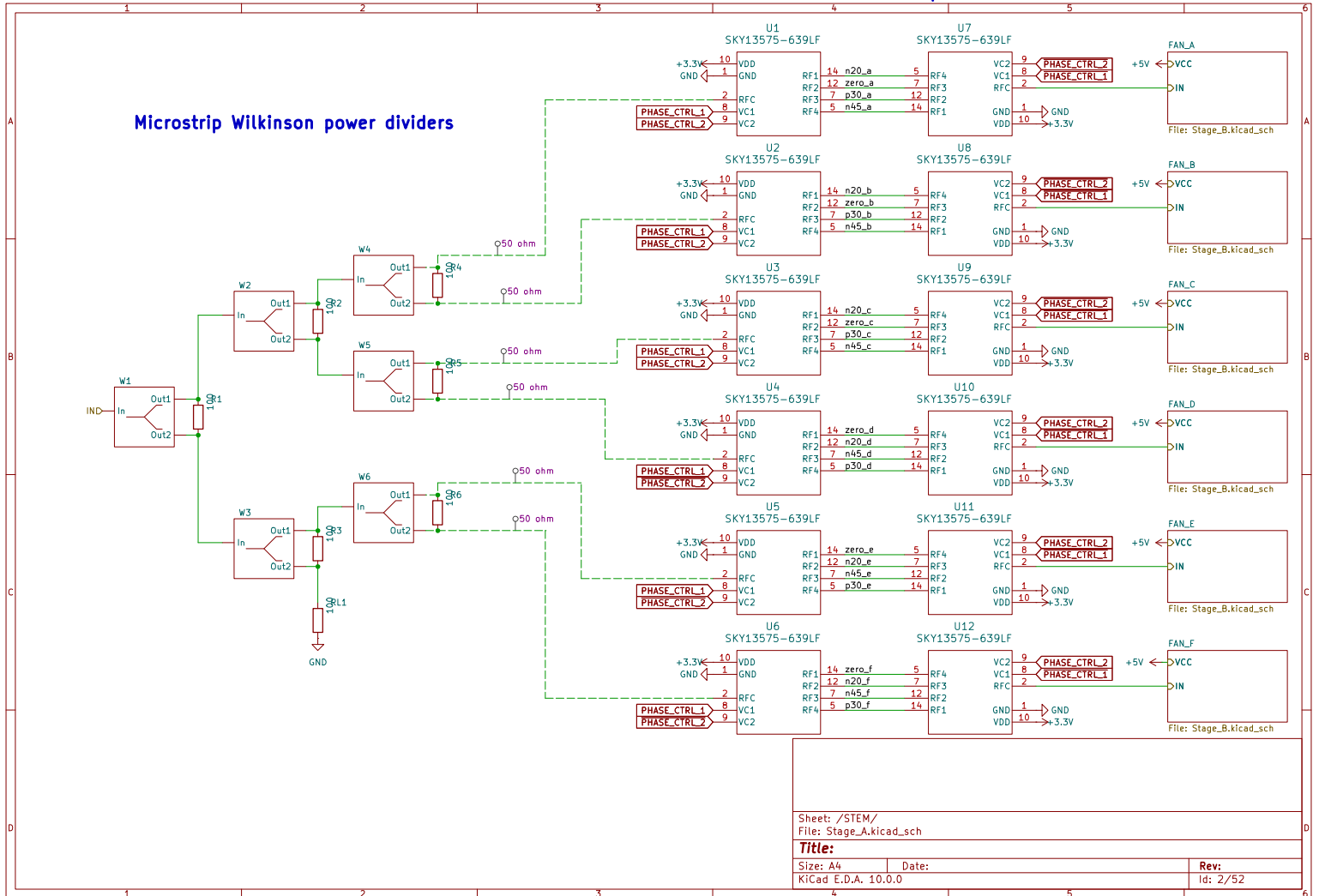




positive offset (mm)				
-45 degrees	-20 degrees	0 degrees	30 degrees	
44.92	31.37	18.61	0.00	
26.95	18.82	11.17	0.00	
8.98	6.27	3.72	0.00	
0.00	2.71	5.26	8.98	
0.00	8.13	15.79	26.95	
0.00	13.55	26.31	44.92	

RF Switches for phase control



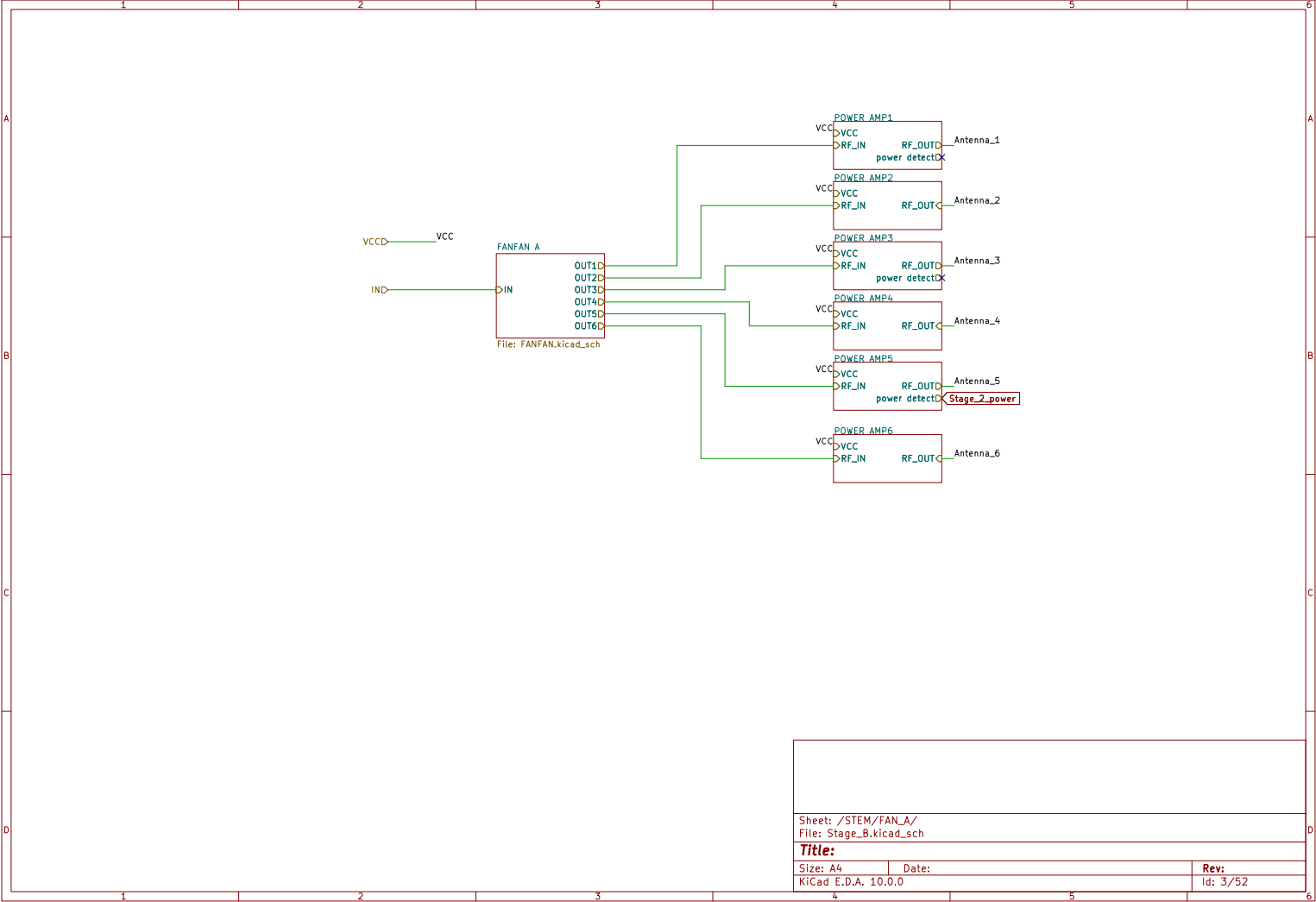
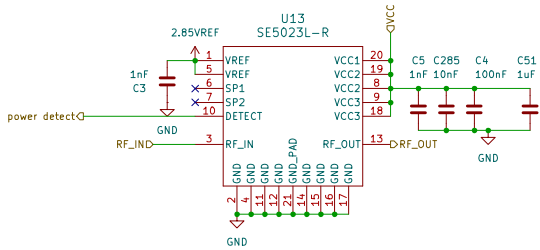


Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_A/POWER AMP1/
File: POWER_AMP.kicad_sch

Title:

Size: A4

Date:

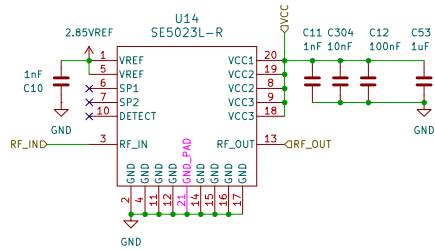
Rev:

KiCad E.D.A. 10.0.0

Id: 4/52

Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_A/POWER AMP2/
File: POWER_AMP2.kicad_sch

Title:

Size: A4

Date:

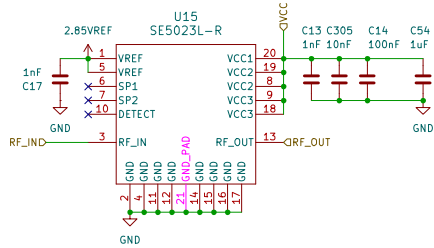
Rev:

KiCad E.D.A. 10.0.0

Id: 5/52

Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_A/POWER AMP4/
File: POWER_AMP2.kicad_sch

Title:

Size: A4

Date:

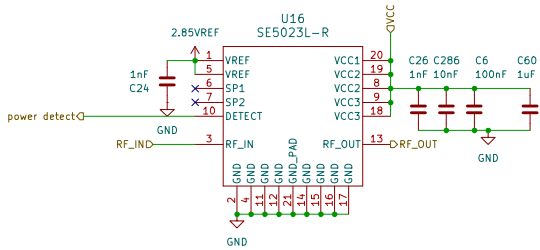
Rev:

KiCad E.D.A. 10.0.0

Id: 6/52

Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_A/POWER AMP3/
File: POWER_AMP.kicad_sch

Title:

Size: A4

Date:

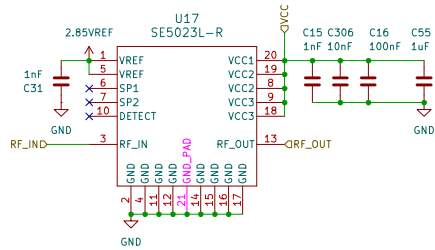
Rev:

KiCad E.D.A. 10.0.0

Id: 7/52

Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_A/POWER AMP6/
File: POWER_AMP2.kicad_sch

Title:

Size: A4

Date:

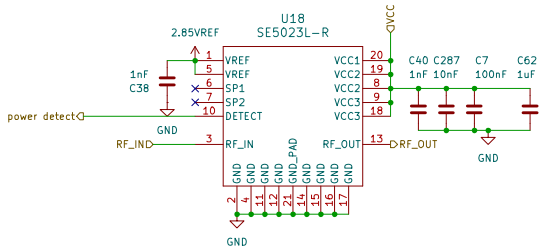
Rev:

KiCad E.D.A. 10.0.0

Id: 8/52

Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_A/POWER AMP5/
File: POWER_AMP.kicad_sch

Title:

Size: A4

Date:

Rev:

KiCad E.D.A. 10.0.0

Id: 9/52

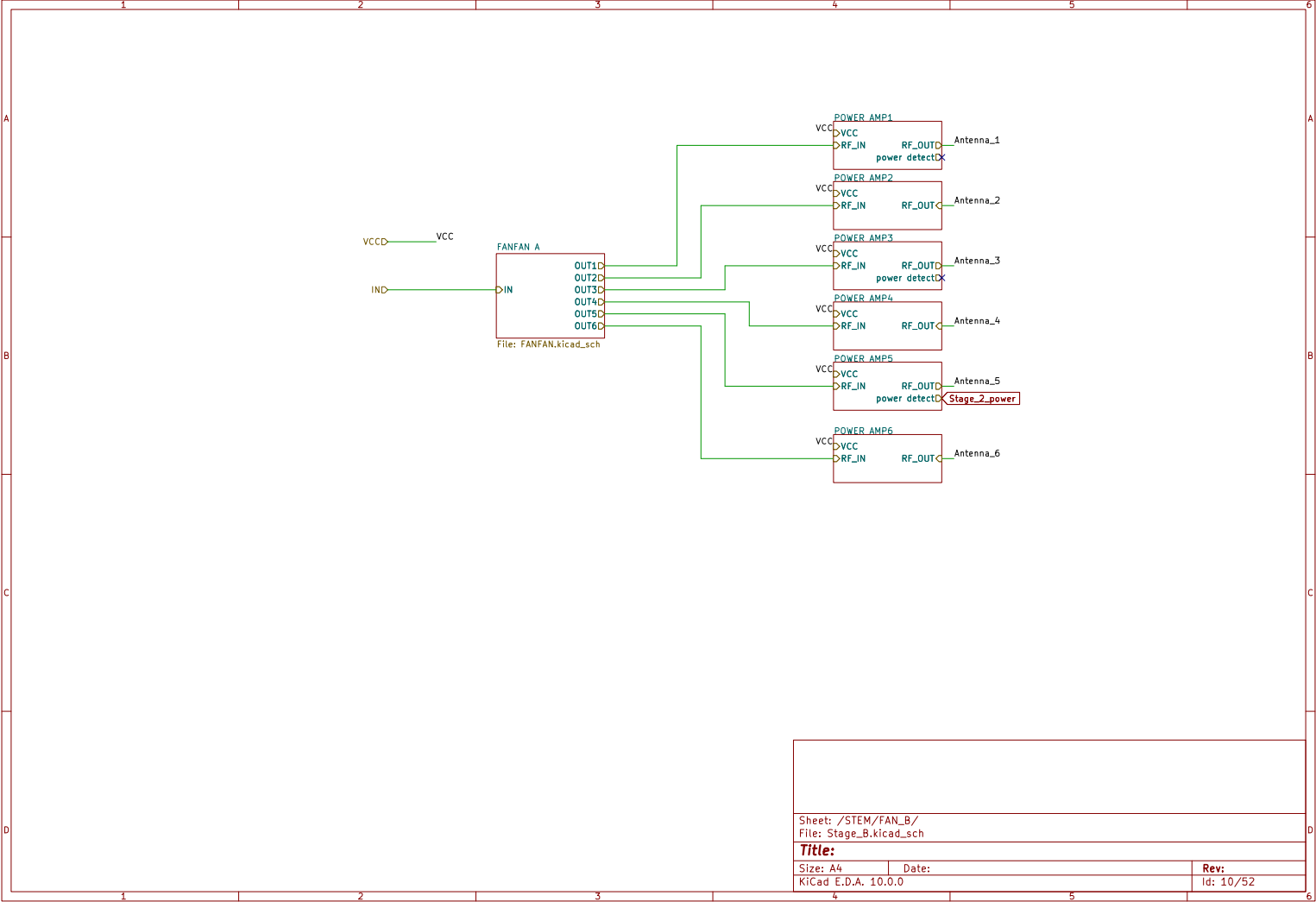
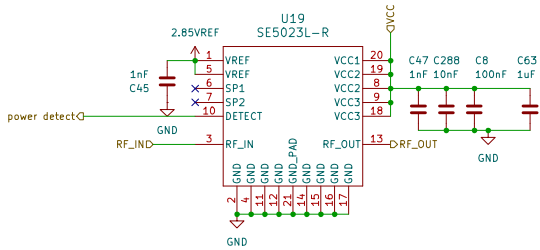


Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_B/POWER AMP1/
File: POWER_AMP.kicad_sch

Title:

Size: A4

Date:

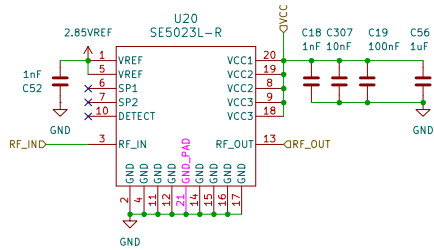
Rev:

KiCad E.D.A. 10.0.0

Id: 11/52

Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_B/POWER AMP2/
File: POWER_AMP2.kicad_sch

Title:

Size: A4

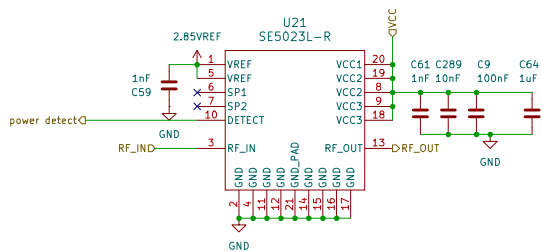
Date:

Rev:

KiCad E.D.A. 10.0.0

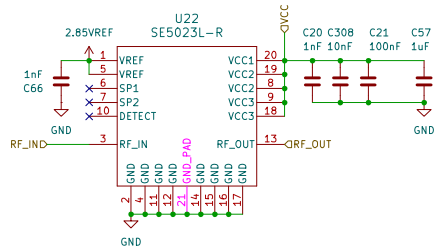
Id: 12/52

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{CASE_MAX}	-40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V



Rev:
Id: 13/52

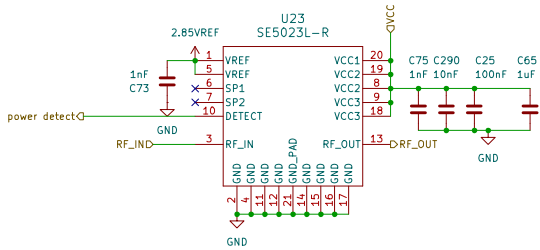
Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{CASE_MAX}	-40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V



Id: 14/52

Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_B/POWER AMP5/
File: POWER_AMP.kicad_sch

Title:

Size: A4

Date:

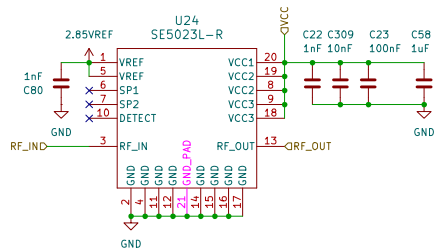
Rev:

KiCad E.D.A. 10.0.0

Id: 15/52

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{CASE_max}	-40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{CASE_max}	-40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V



Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 16/52

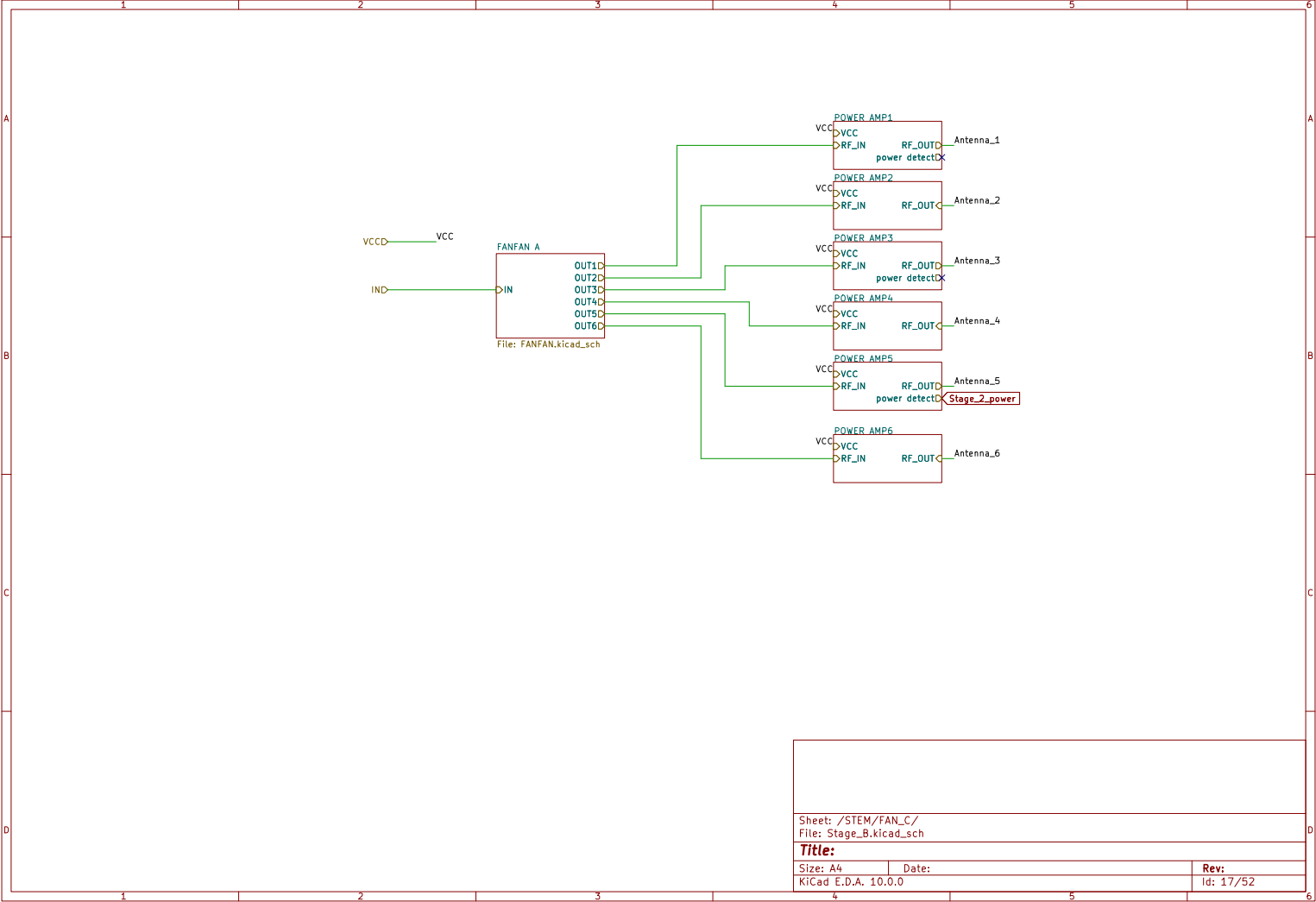
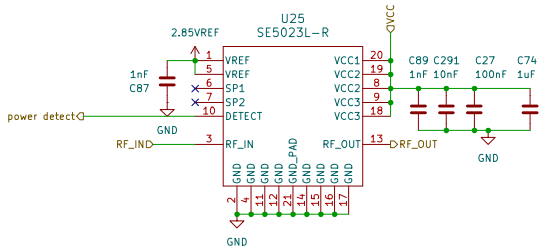


Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_C/POWER AMP1/
File: POWER_AMP.kicad_sch

Title:

Size: A4

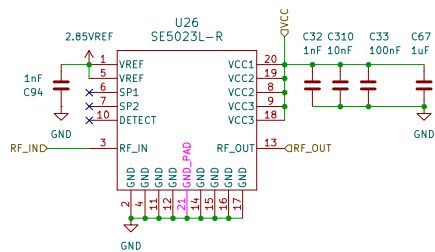
Date:

Rev:

KiCad E.D.A. 10.0.0

Id: 18/52

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{CASE_max}	−40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V

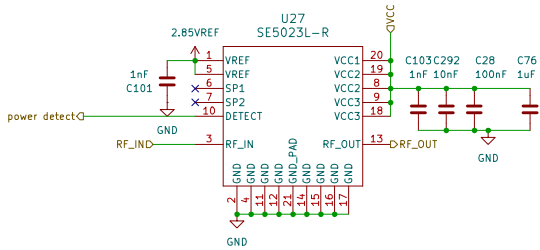


Size: A4	
KiCad E.D.A. 10.0.0	

Id: 19/52

Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_C/POWER AMP3/
File: POWER_AMP.kicad_sch

Title:

Size: A4

Date:

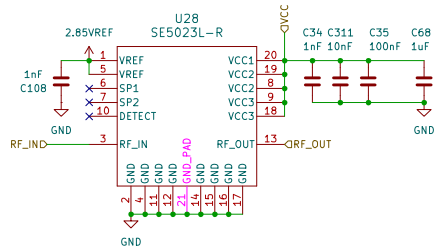
Rev:

KiCad E.D.A. 10.0.0

Id: 20/52

Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_C/POWER AMP6/
File: POWER_AMP2.kicad_sch

Title:

Size: A4

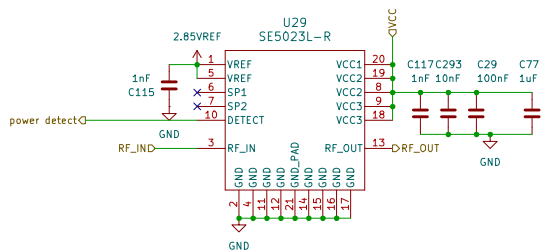
Date:

Rev:

KiCad E.D.A. 10.0.0

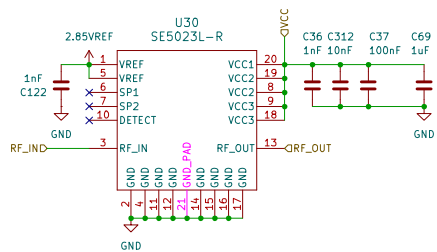
Id: 21/52

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{CASE_max}	-40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V

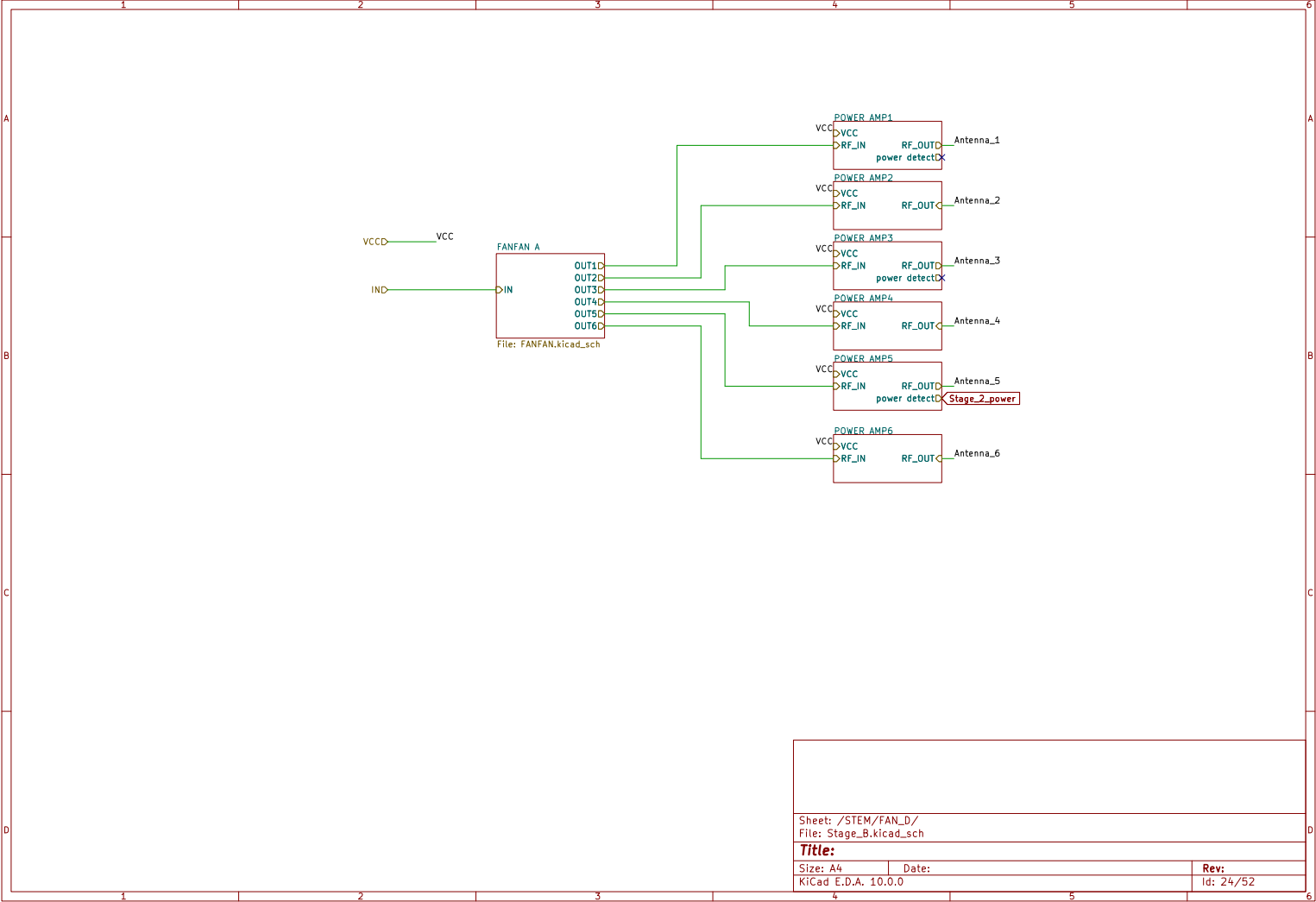


Id: 22/52

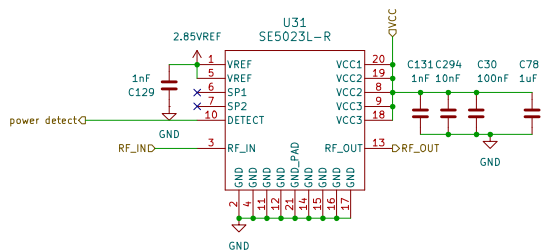
Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{CASE_MAX}	-40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V



Id: 23/52



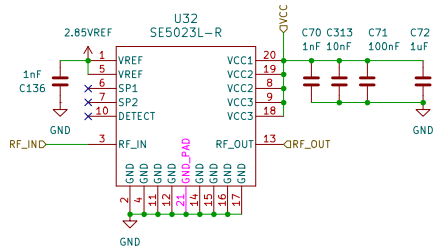
Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{CASE_max}	-40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V



Id: 25/52

Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_D/POWER AMP2/
File: POWER_AMP2.kicad_sch

Title:

Size: A4

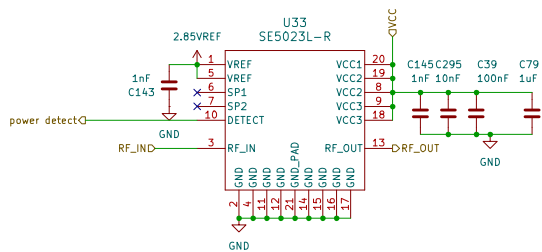
Date:

Rev:

KiCad E.D.A. 10.0.0

Id: 26/52

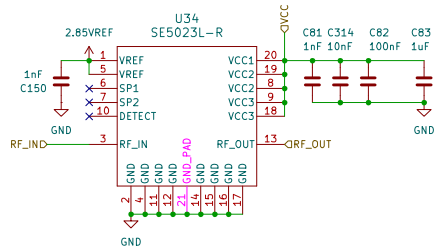
Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{case_max}	-40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V



Id: 27/52

Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_D/POWER AMP6/
File: POWER_AMP2.kicad_sch

Title:

Size: A4

Date:

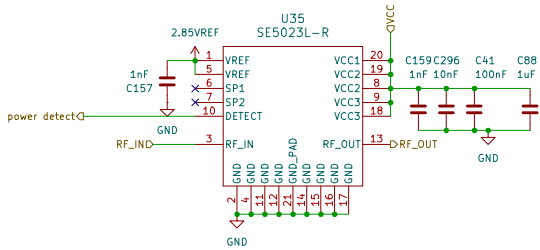
Rev:

KiCad E.D.A. 10.0.0

Id: 28/52

Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_D/POWER AMP5/
File: POWER_AMP.kicad_sch

Title:

Size: A4

Date:

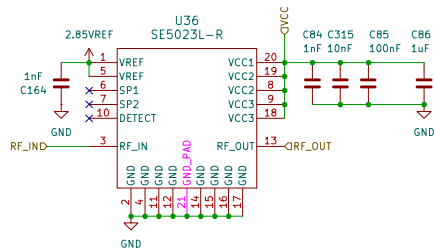
Rev:

KiCad E.D.A. 10.0.0

Id: 29/52

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{CASE_max}	-40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{CASE_max}	-40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V



Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 30/52

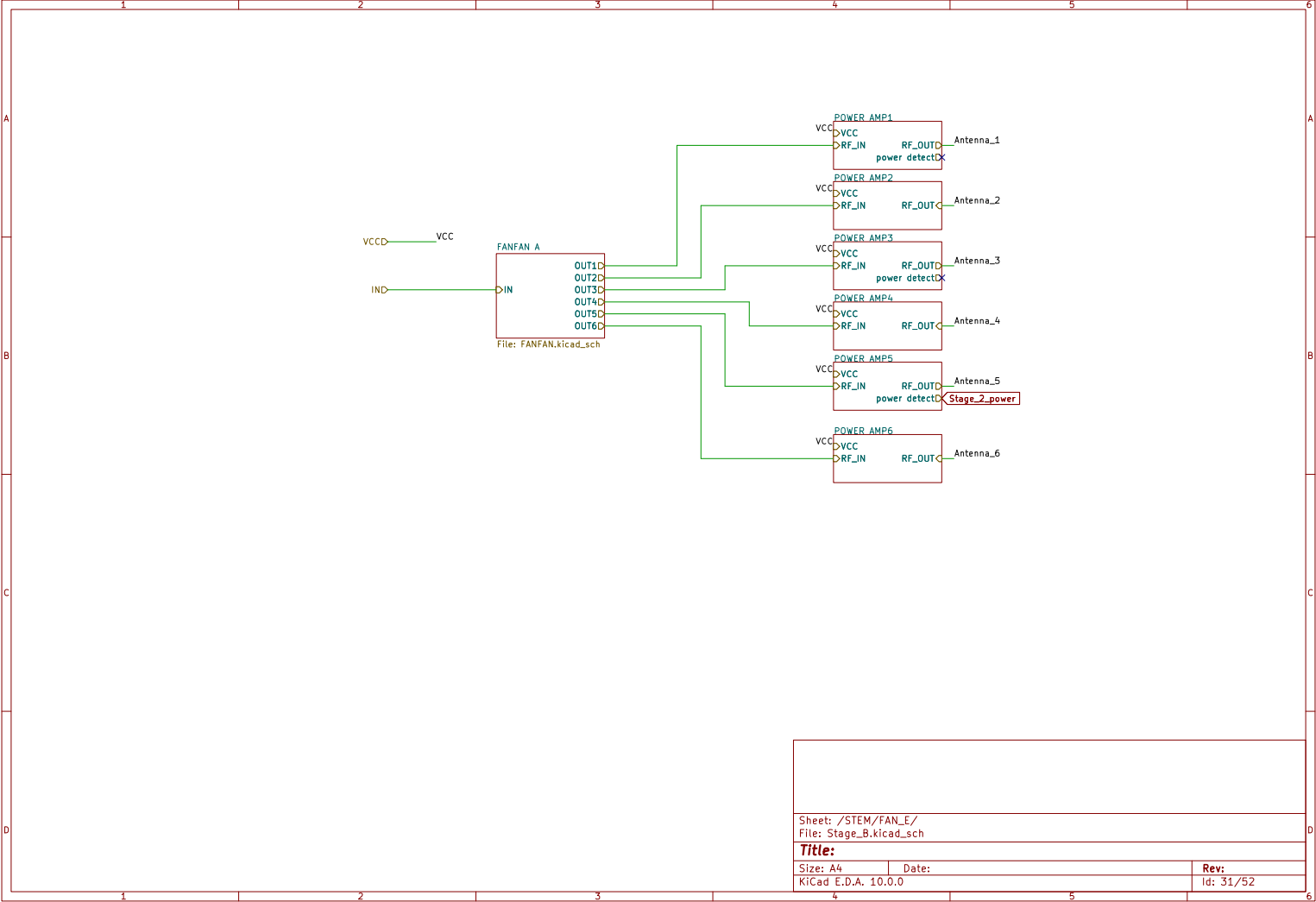
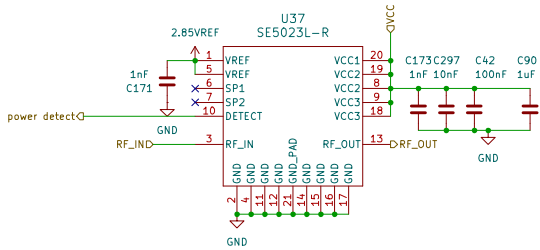


Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_E/POWER AMP1/
File: POWER_AMP.kicad_sch

Title:

Size: A4

Date:

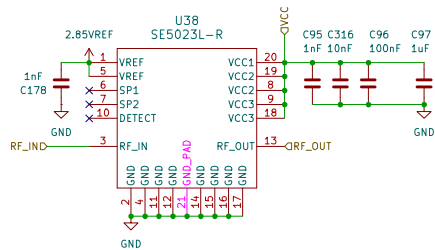
Rev:

KiCad E.D.A. 10.0.0

Id: 32/52

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{CASE_max}	-40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V

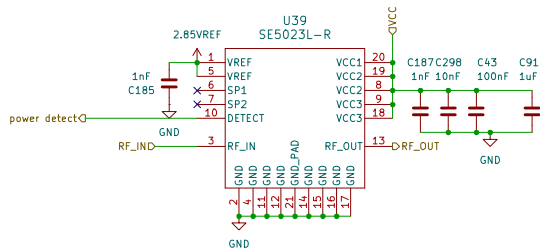
Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{CASE_max}	-40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V



Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 33/52

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{case_max}	-40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V

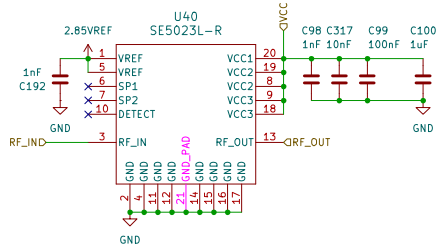
Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{case_max}	-40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V



Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 34/52

Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_E/POWER AMP6/
File: POWER_AMP2.kicad_sch

Title:

Size: A4

Date:

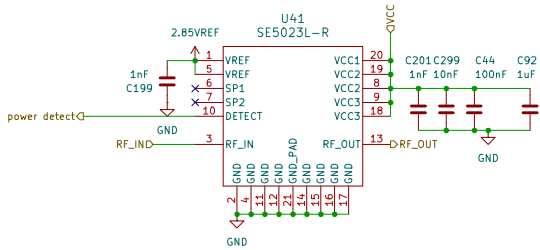
Rev:

KiCad E.D.A. 10.0.0

Id: 35/52

Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_E/POWER AMP5/
File: POWER_AMP.kicad_sch

Title:

Size: A4

Date:

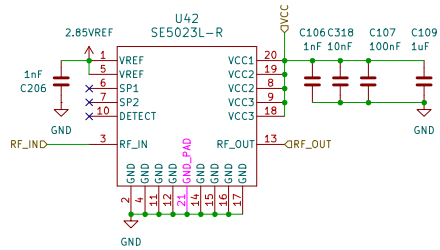
Rev:

KiCad E.D.A. 10.0.0

Id: 36/52

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{CASE_max}	-40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{CASE_max}	-40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V



Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 37/52

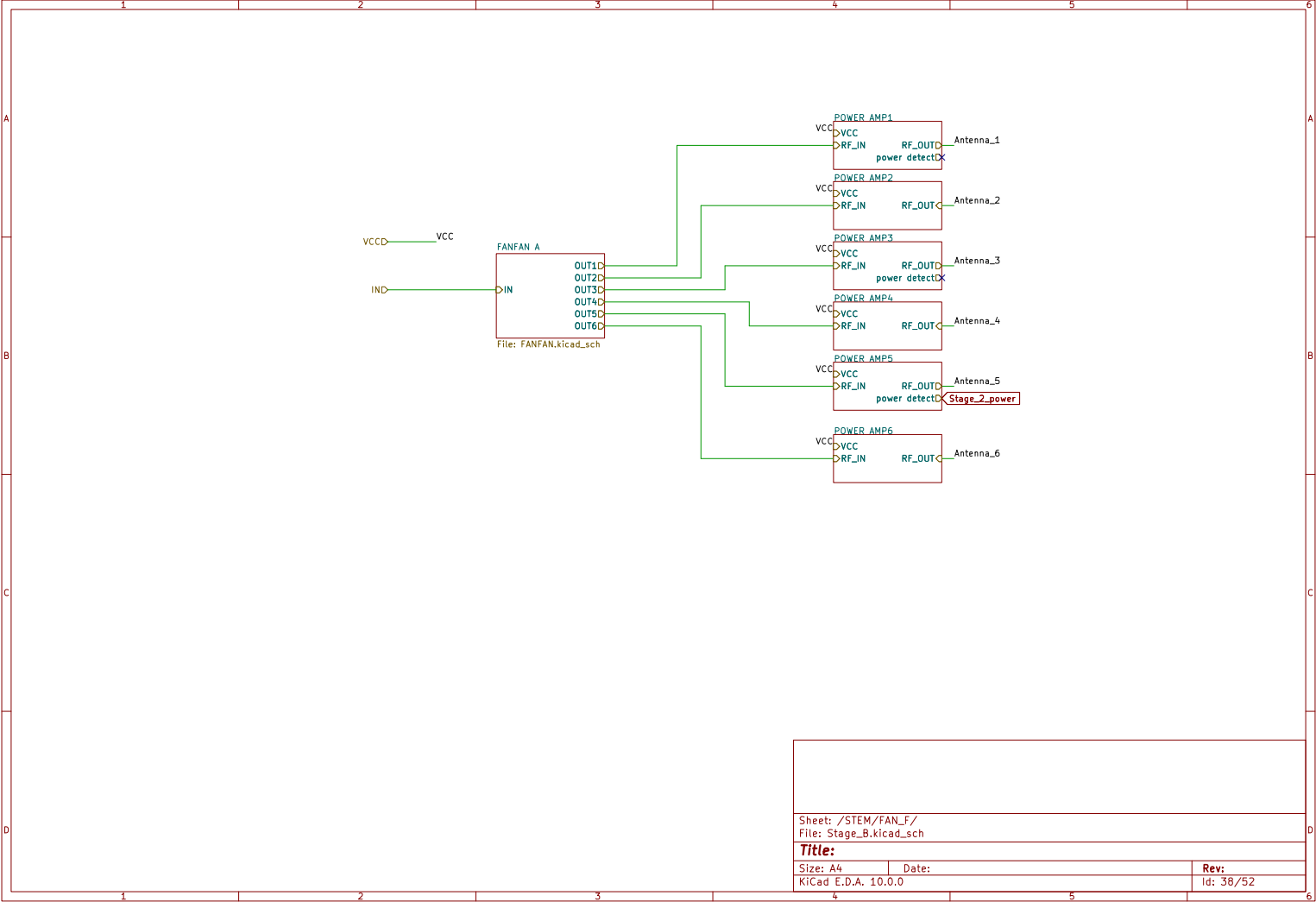
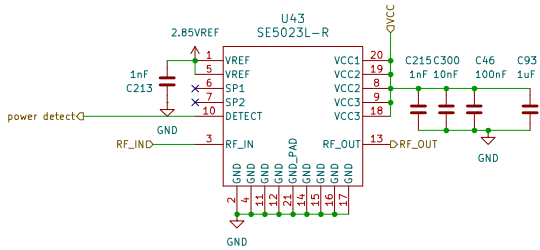


Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_F/POWER AMP1/
File: POWER_AMP.kicad_sch

Title:

Size: A4

Date:

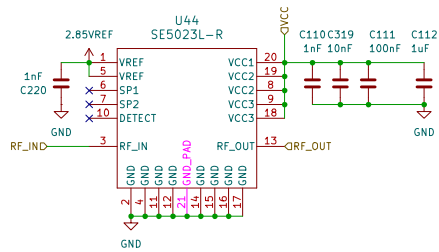
Rev:

KiCad E.D.A. 10.0.0

Id: 39/52

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{CASE_max}	-40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V

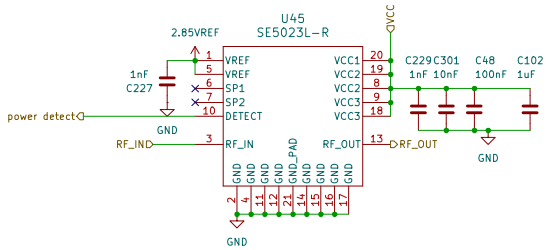
Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{CASE_max}	-40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V



Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 40/52

Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_F/POWER AMP3/
File: POWER_AMP.kicad_sch

Title:

Size: A4

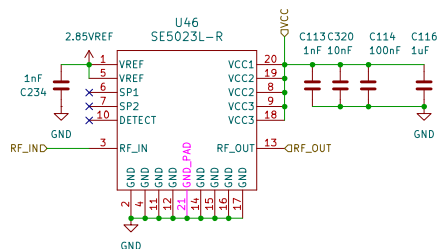
Date:

Rev:

KiCad E.D.A. 10.0.0

Id: 41/52

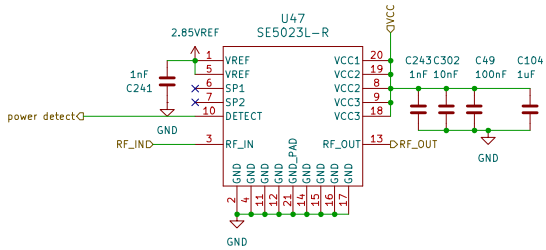
Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{CASE_max}	-40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V



Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 42/52

Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /STEM/FAN_F/POWER AMP5/
File: POWER_AMP.kicad_sch

Title:

Size: A4

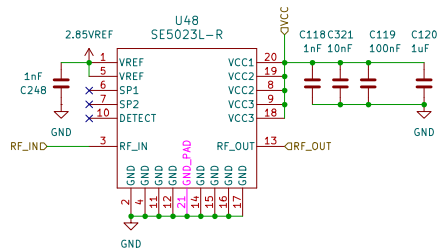
Date:

Rev:

KiCad E.D.A. 10.0.0

Id: 43/52

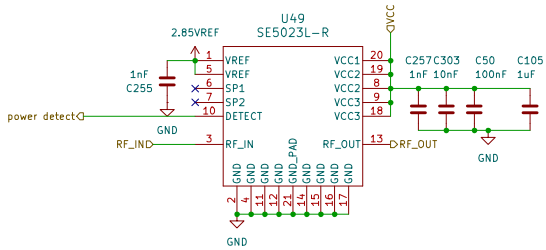
Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	T _{CASE_max}	-40	+85	°C
Reference voltage	V _{REF}	2.8	2.9	V



Id: 44/52

Table 3. SE5023L Recommended Operating Conditions

Parameter	Symbol	Minimum	Maximum	Units
Supply voltage VCC3	VCC	3.0	5.5	V
Supply voltage VCC1, VCC2		3.0	VCC3	V
Maximum case temperature	Tcase_max	-40	+85	°C
Reference voltage	VREF	2.8	2.9	V



Sheet: /POWER AMP6/
File: POWER_AMP.kicad_sch

Title:

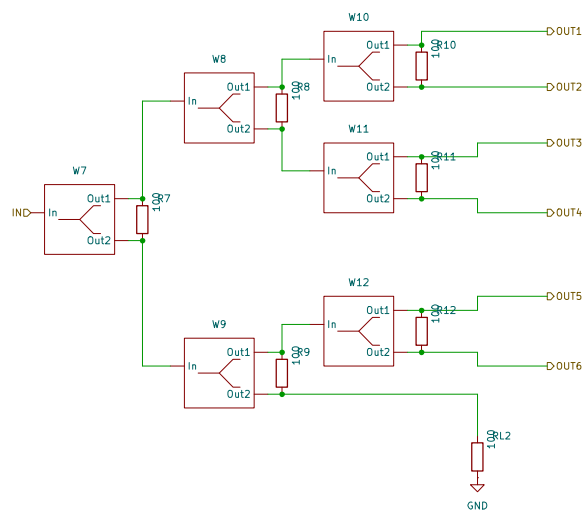
Size: A4

Date:

Rev:

KiCad E.D.A. 10.0.0

Id: 45/52



Sheet: /STEM/FAN_A/FANFAN A/
File: FANFAN.kicad_sch

Title:

Size: A4
KiCad E.D.A. 10.0.0

Date:

Rev:

Id: 46/52

Id: 47/52





Size: A4	
KiCad E.D.A. 10.0.0	

Rev:

Id: 49/52

Id: 50/52



Size: A4	
KiCad E.D.A. 10.0.0	

Rev:
Id: 51/52

Id: 52/52